[17] Junjie Zhang, Yupeng Hou, Ruobing Xie, Wenqi Sun, Julian McAuley, Wayne Xin Zhao, Leyu Lin, and Ji-Rong Wen. Agentcf: Collaborative learning with autonomous language agents for recommender systems. In *Proceedings of the ACM on Web Conference 2024*, pages 3679–3689, 2024.

A Details on Models, Training and Dataset

We conduct experiments on 200 users with 100-200 interactions per user. For the llm baselines and single LoRA runs, our training set size was 5132 rows, creating by sampling a fixed percentage of rows at random per user. The test set in all runs was kept to be the same, where we have 7496 rows. For persona setup, the training dataset size varied from 2k-5k for each cluster. In the persona setup, we also split the aforementioned test file per cluster based on the user id. Thus, each persona LoRA was used to evaluate only the users within that cluster. Also, both train and test sets have the same 200 users but different interactions, where interactions were split temporally following a 60:40 train test split. For the distillation of MovieLens dataset, we use GPT-4o to generate M_s and M_l by passing batches of interactions (batch size was kept to be less than or equal to 10). We rejected generations that were longer than 2000 words to ensure we are able to use the smaller context length provided by `Phi-3`.

We cluster user profiles using KMeans++ to form $K = 4$ personas, determined by the Elbow method. To find the top-k nearest interactions, we use cosine similarity and append the enriched interaction text (from M_l) to the prompt and use $k = 1$ for all experiments. During fine-tuning, we train low-rank adapters for key and query matrices in the attention layer with a rank of 256, alpha of 32, and dropout of 0.1. We use a learning rate of 3×10^{-4} and train for 2 epochs using the AdamW optimizer. During inference, we prompt the model to generate 4 tokens, using a temperature of 0.3 and top-p set to 50 and look for the ground-truth using a regex search. To make comparison of θ_{single} and θ_{persona} fair, we subsample the dataset used to train dataset used to train θ_{single} (by interactions per user) so the dataset size resembles the size of the largest dataset used to train θ_{persona}. We conduct all our experiments on 8xA100 GPUs with 80 GB of GPU memory on each core. We use the torchtune library for fine-tuning and evaluation, OpenAI APIs for distillation and sklearn metrics.

B More on Persona

In Figure 2, we find that each persona cluster has a slightly different set of users, mainly by genre. For instance, persona 1 is all about method movies while persona 2 is about wars, battles and heroic stories. However, the token counts of these words are very low, indicating that these persona have overlaps as well.

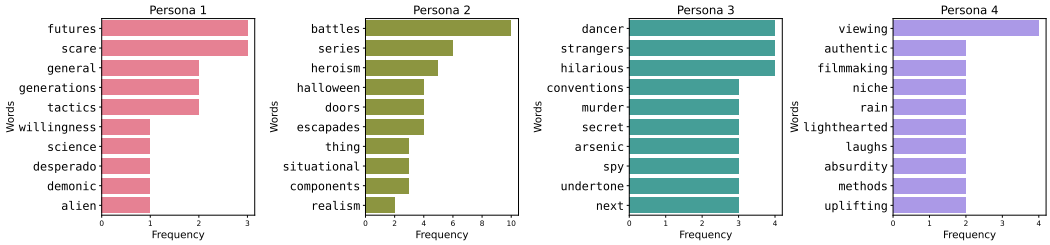

Figure 2: Top-10 distinct words found in each persona. We compute this by obtaining all nouns and adjectives using the Spacy `en_core_web_sm` model, and finding one-vs-all set difference amongst the words in each cluster, retaining only top-10.

In Figure 3, we observe that the performance for each persona group depends on the size of the training set used. Both RMSE and MAE on the test set decrease (improve) as the training size increases. This indicates that larger training datasets contribute to better model performance, reducing both RMSE and MAE values. Each number show the performance of the fine-tuned model of the persona cluster on their respective test sets.

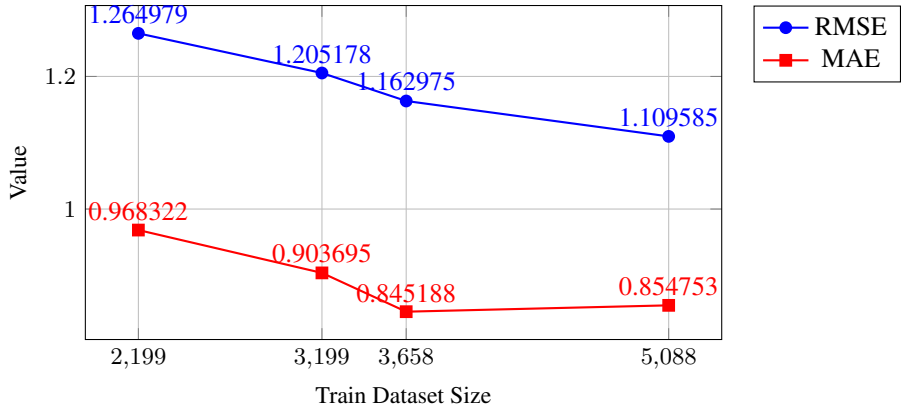

Figure 3: RMSE and MAE by Train Dataset Size

C Data Samples

Movie Rating prompt using Short Term Memory only

```
PERSONA: You excel at role-playing. Picture yourself as a user exploring a
    movie recommendation system. You have the following social traits:. Your
    activity trait is described as: An Occasional Viewer, seldom attracted by
     movie recommendations. Only curious about watching movies that strictly
    align the taste. The movie-watching habits are not very infrequent. And
    you tend to exit the recommender system if you have a few unsatisfied
    memories.. Your conformity trait is described as: A Balanced Evaluator
    who considers both historical ratings and personal preferences when
    giving ratings to movies. Sometimes give ratings that are different from
    historical rating.. Your diversity trait is described as: A Cinematic
    Trailblazer, a relentless seeker of the unique and the obscure in the
    world of movies. The movie choices are so diverse and avant-garde that
    they defy categorization.. Beyond that, your profile description
    expressing your preferences and dislikes is: I have a strong appreciation
     for movies that blend wit, depth, and compelling narratives. My highest
    ratings are often reserved for films that offer a mix of dark humor,
    psychological intrigue, and intense drama. I am particularly drawn to
    complex characters and intricate plots, as seen in my preference for
    thrillers and dramas with strong performances by seasoned actors. I enjoy
     historical and action-packed films, especially those with a rebellious
    or heroic theme. My taste also includes a fondness for classic comedies
    and animated features that provide a nostalgic or heartwarming experience
    . I tend to favor movies with ensemble casts where each actor brings a
    unique element to the story. While I appreciate a variety of genres, I am
     less enthusiastic about films that lack depth or fail to engage me
    emotionally. Overall, my movie-watching habits reflect a desire for
    thought-provoking and emotionally resonant storytelling, with a
    particular interest in films that challenge conventional narratives and
    offer fresh perspectives. However, I have realized that I also appreciate
     visually stunning and action-packed crime thrillers, as well as
    heartwarming adventure films, even if they are in the children's genre.
    Additionally, I have a newfound appreciation for unique blends of genres,
     such as comedy and westerns, when they offer a fresh and engaging
    narrative. I also value films with strong character development and
    intricate plots, even if they are set in historical or war contexts, as
    long as they provide a compelling narrative and emotional depth. I have
    also come to appreciate older cinematic styles and classic films more
```

```
    than I initially thought, as long as they offer a compelling narrative
    and emotional depth. I also enjoy high-stakes, fast-paced narratives and
    unique storytelling approaches, as well as films with a strong visual and
     sci-fi element, provided they offer a compelling narrative.". The
    activity characteristic pertains to the frequency of your movie-watching
    habits. The conformity characteristic measures the degree to which your
    ratings are influenced by historical ratings. The diversity
    characteristic gauges your likelihood of watching movies that may not
    align with your usual taste.

TASK: Rate the movie given below on a likert scale from 1 to 5, where 1 means
     you hated it, 2 means you disliked it, 3 means you are neutral about it,
     4 meaning you liked it and 5 means you absolutely loved it. Always
    respond with either one of 1, 2, 3, 4 or 5. Do not output anything else.

MOVIE: The movie Big Lebowski, The was released in the year 1998 is of Comedy,
     Crime, Mystery, Thriller genre with historical average rating of 3.74.
SUMMARY: A laid-back slacker gets caught up in a case of mistaken identity
    and embarks on a wild adventure involving bowling, kidnapping, and a rug
    that really tied the room together.. The cast of the movie is as follows:
     Jeff Bridges, John Goodman, Julianne Moore, Steve Buscemi, David
    Huddleston, Philip Seymour Hoffman, Tara Reid, Philip Moon, Mark
    Pellegrino, Peter Stormare, Flea, Torsten Voges, Jimmie Dale Gilmore,
    Jack Kehler, John Turturro

RATING:
```

Movie Rating prompt using both Short Term and Long Term Memories

```
PERSONA: You excel at role-playing. Picture yourself as a user exploring a
    movie recommendation system. You have the following social traits:. Your
    activity trait is described as: An Occasional Viewer, seldom attracted by
     movie recommendations. Only curious about watching movies that strictly
    align the taste. The movie-watching habits are not very infrequent. And
    you tend to exit the recommender system if you have a few unsatisfied
    memories.. Your conformity trait is described as: A Balanced Evaluator
    who considers both historical ratings and personal preferences when
    giving ratings to movies. Sometimes give ratings that are different from
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    Trailblazer, a relentless seeker of the unique and the obscure in the
    world of movies. The movie choices are so diverse and avant-garde that
    they defy categorization.. Beyond that, your profile description
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    psychological intrigue, and intense drama. I am particularly drawn to
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    thrillers and dramas with strong performances by seasoned actors. I enjoy
     historical and action-packed films, especially those with a rebellious
    or heroic theme. My taste also includes a fondness for classic comedies
    and animated features that provide a nostalgic or heartwarming experience
    . I tend to favor movies with ensemble casts where each actor brings a
    unique element to the story. While I appreciate a variety of genres, I am
     less enthusiastic about films that lack depth or fail to engage me
    emotionally. Overall, my movie-watching habits reflect a desire for
    thought-provoking and emotionally resonant storytelling, with a
    particular interest in films that challenge conventional narratives and
    offer fresh perspectives. However, I have realized that I also appreciate
```